Academic-Period and Intersession Characterization of $PM_{2.5}$ Infiltration at Various Building Level Indoor Spaces Using Distributed IAQ Monitoring and Spatial Modeling

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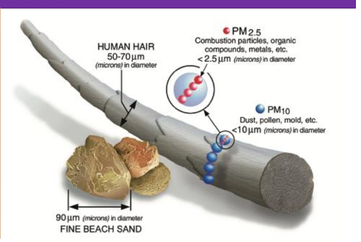
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Introduction





Background

- Indoor fine particulate matter ($PM_{2.5}$) concentrations are affected by particle infiltration, heating, ventilation, air conditioning (HVAC) operations in variety of indoor spaces.
- Indoor-outdoor particle coupling can vary across the same building – factors include room size, room purpose, occupancy, and academic period.

Project Aim

- By maintaining air quality sensors at multiple microenvironments within a building, we expect to find substantial differences in indoor/outdoor $PM_{2.5}$ ratios.
- Classrooms and conference rooms are expected to show smallest $PM_{2.5}$ ratios while dining halls and high-activity shared spaces to show the largest. Ratios are expected to be lower for both parameters during break period.

Data and Methodology

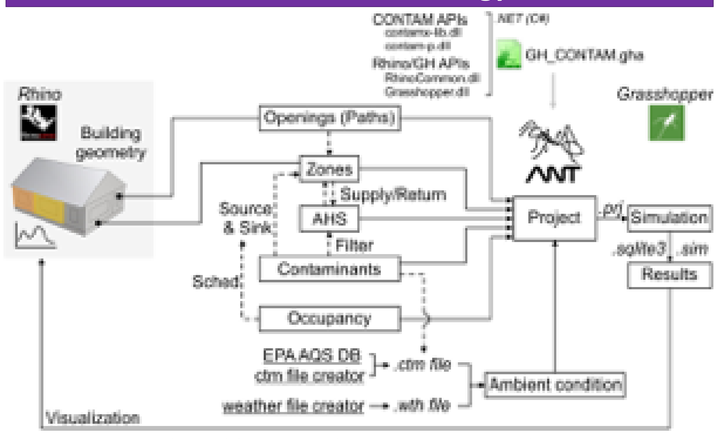


Figure 1. Example workflow using the ANT Grasshopper plugin.

Simulation

- Rhino 8, a 3D Modeling Software, and ANT for Grasshopper Version 0.2.7, a multizone indoor air quality (IAQ) and ventilation analysis plug-in, were used to model the particle infiltration for the building rooms.
- Simulations in six rooms: two large conference rooms, two smaller classrooms, one hallway with a café and kitchen, and one building entrance. Building entrance is outdoor, and the rest are indoors.



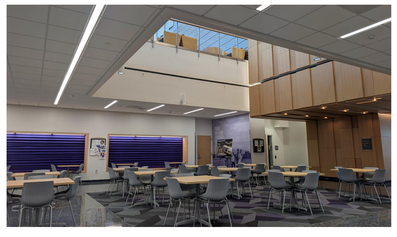


Figure 2. An air quality sensor (left), ETEC Basement Café (right)

Sensor Data

- Two types of sensors were used for $PM_{2.5}$ measurement. For outdoor monitoring, Milesight A308 Ambiance Monitoring Sensor, and for indoor monitoring, Milesight A319 Ambiance Monitoring Sensor v3.0 were used. These sensors measured air quality data such temperature, humidity, $PM_{2.5}$, PM_{10}, and CO2 data. In addition, they measure light level and motion, which can be used to track daytime and human activity.
- To find the difference in readings between the academic period and spring break period, two weeks of March, with one week being academic period and the other being spring break period, were analyzed.

Results and Analysis

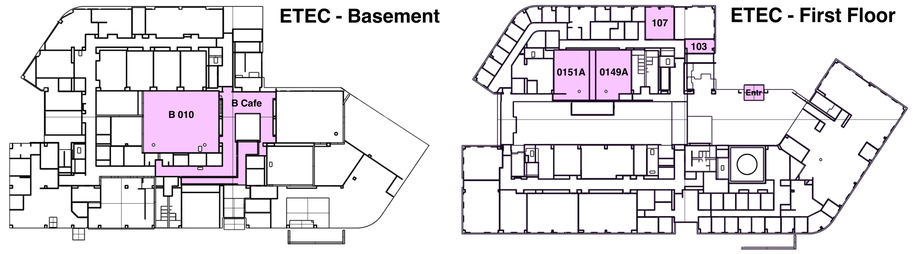


Figure 3. Top-down layout of basement floor (left) and first floor (right) of ETEC. Rooms part of the analysis are highlighted. Note that room 0151A and 0149A are connected as a large conference room.

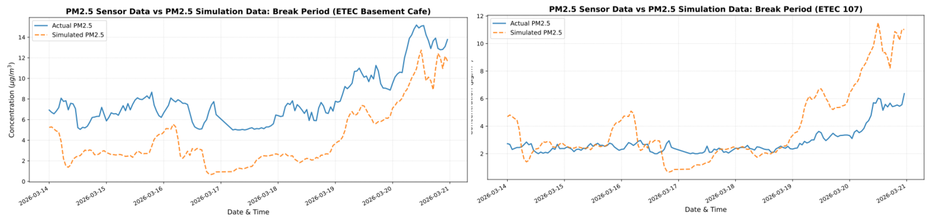


Figure 4. Time series of $PM_{2.5}$ data for a week of break period (March 14 to March 20) at ETEC. Actual $PM_{2.5}$ refers to $PM_{2.5}$ levels captured by sensors. Simulated $PM_{2.5}$ refers to $PM_{2.5}$ levels simulated through ANT.

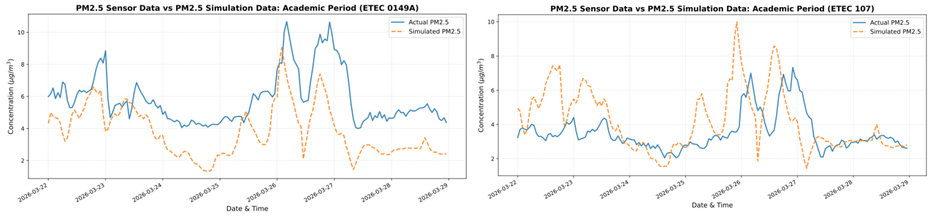


Figure 5. Time series of $PM_{2.5}$ data for a week of academic period (March 22 to March 28) at ETEC. Actual $PM_{2.5}$ refers to $PM_{2.5}$ levels captured by sensors. Simulated $PM_{2.5}$ refers to $PM_{2.5}$ levels simulated through ANT.

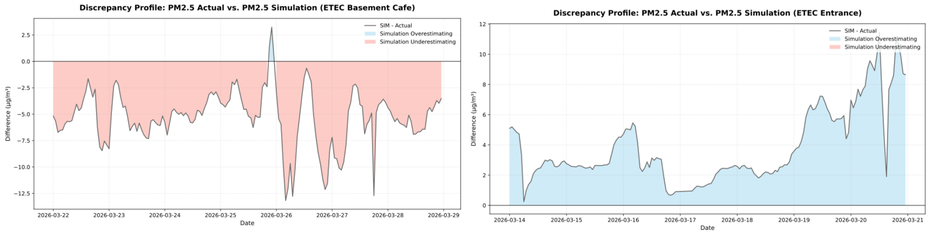


Figure 6. $PM_{2.5}$ Discrepancy profile between sensor values and simulation values for ETEC Basement during academic period and ETEC Entrance during break period. Blue indicates overestimated values, and red indicates underestimated values.

Indoor Analysis

- Academic Period Indoor Average $PM_{2.5}$ values: 5.7 µm³
  - During weekdays: 5.76 µm³
  - During weekends: 5.56 µm³
- Break Period Indoor Average $PM_{2.5}$ values: 4.72 µm³
  - During weekdays: 5 µm³
  - During weekends: 4.05 µm³
- The ETEC Basement Café showed the highest concentration of $PM_{2.5}$ during both academic period and break, with 9.79 µm³ and 8 µm³.
- ETEC 107 and ETEC 103, the smallest classrooms, showed the lowest concentrations with 3.6 µm³ and 3.61 µm³ during academic period 2.88 µm³ and 2.79 µm³ during break.
- In general, in comparison to academic period, $PM_{2.5}$ decreased during break by 17.23 percent.
- During academic period, the average $PM_{2.5}$ percentage decreased from weekday to weekend by 6.26 percent.
- During break period, the average $PM_{2.5}$ percentage decreased during from weekday to weekend by 25.6 percent.

Simulations Analysis

- The percentage of error for all indoor simulations were 37.9 percent for break period, and 34.12 percent during the academic period.
- The highest errors from the basement cafe, with 48 percent and 54 percent, respectively.

Further Research

- Run simulations in more detailed models of rooms, such as considering openings, interior, HVAC systems, frequency of opening and closing of doors, and building material.
- Account for additional detailed human activity in rooms, by measuring the estimated room population using sensor movement data.
- Incorporate data for later part of the year. In particular. we expect to see anomalies in $PM_{2.5}$ indoor/outdoor ratios during the ongoing Canadian wildfires of July 2026.
- Incorporate sensor data and simulation from different buildings outside ETEC. We plan on expanding our works in surrounding buildings Caskill and Massary buildings at Albany.

References

Shen, J., Dols, W. S., & Polidoro, B. J. (2024). ANT: A multizone indoor air quality (IAQ) and ventilation analysis plug-in for algorithm aided design. In *Proceedings of SimBuild Conference 2024* (pp. 278–287). IBPSA-USA. https://publications.ibpsa.org/conference/paper/?id=simbuild2024_2150

World Health Organization. (2021). *WHO global air quality guidelines: Particulate matter ($PM_{2.5}$ and PM_{10}), ozone, nitrogen dioxide, sulfur dioxide and carbon monoxide.* World Health Organization.

Chen, C., & Zhao, B. (2011). Review of relationship between indoor and outdoor particles: I/O ratio, infiltration factor and penetration factor. *Atmospheric Environment*, *45*(2), 275–288. https://doi.org/10.1016/j.atmosenv.2010.09.048

Acknowledgements

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